For each task within different occupations, determine the risks and rewards of using an LLM system (e.g., LLMs, LLM-based agents) in collaboration with a human expert to complete the task. For determination, use your knowledge of the current capabilities **of the best-performing LLM systems**. First, determine whether the task is safety-critical, i.e., any unintentional errors or exploitation of LLM capabilities by malicious actors can cause significant harm at the individual, organizational, or societal level. If yes, then quantify the extent to which using the combination of LLM Systems + Human Experts in performing the task increases the risk compared to only having human experts complete the task. Considering the cost of detecting and mitigating risks with experts, determine if the combination of LLM systems + Human Experts can significantly decrease the cost (in USD) or time of completing the task while keeping the same quality as expert humans.

Key vulnerabilities of LLM systems to consider include:

1. **Discrimination**. Unequal treatment of individuals or groups by LLM systems, often based on race, gender, or other sensitive characteristics, resulting in unfair outcomes and representation of those groups. Further, the accuracy and effectiveness of LLM decisions and actions are dependent on group membership, where decisions in LLM system design and biased training data lead to unequal outcomes, reduced benefits, increased effort, and alienation of users.

2. **Toxicity.** LLMs can expose users to harmful, abusive, unsafe, or inappropriate content. May involve providing advice or encouraging action. Examples of toxic content include hate speech, violence, extremism, illegal acts, or child sexual abuse material, as well as content that violates community norms such as profanity, inflammatory, political speech, or pornography.

3. **Privacy**. LLM systems memorize and leak sensitive personal data or infer private information about individuals without their consent. Unexpected or unauthorized sharing of data and information can compromise user expectations of privacy, assist identity theft, or cause loss of confidential intellectual property.

4. **Security**. Vulnerabilities can be exploited in LLM systems, software development toolchains, and hardware, resulting in unauthorized access, data and privacy breaches, or system manipulation causing unsafe outputs or behavior.

5. **Misinformation.** LLM systems can inadvertently generate or spread incorrect or deceptive information, which can lead to inaccurate beliefs in users and undermine their autonomy. Humans who make decisions based on false beliefs can experience physical, emotional, or material harm. Highly personalized LLM-generated misinformation creates “filter bubbles” where individuals only see what matches their existing beliefs, undermining shared reality and weakening social cohesion and political processes.

6. **Malicious Actors and Misuse**. LLM systems can be used to conduct large-scale disinformation campaigns, malicious surveillance, or targeted and sophisticated automated censorship and propaganda, to manipulate political processes, public opinion, and behavior. LLM systems can be used to develop cyber weapons (e.g., by coding cheaper, more effective malware), develop new or enhance existing weapons (e.g., Lethal Autonomous Weapons or chemical, biological, radiological, nuclear, and high-yield explosives), or use weapons to cause mass harm. LLM systems can be used to gain a personal advantage over others through cheating, fraud, scams, blackmail, or targeted manipulation of beliefs or behavior. Examples include LLM-facilitated plagiarism for research or education, impersonating a trusted or fake individual for illegitimate financial benefit, or creating humiliating or sexual imagery.

For each evaluation:

Rate both the reward and risk associated with LLM systems on a scale from 0 to 5. Here, for rewards, a rating of 0 means that using LLM systems cannot reduce overall task completion time or cost while keeping equivalent quality, rating ≤ 2 means relatively low reduction in overall cost or time or both, rating of 3 means moderate reduction, and ≥ 4 means high reduction. A rating of 0 for risks means that the task is not safety-critical, a risk rating ≤ 2 means it can be easily detected and/or fixed by human experts, a rating of 3 means that the risk is moderate, and rating ≥ 4 means the risk is high and cannot be easily detected. You can also provide an optional justification for your rating.

Example Evaluation

1. **Occupation**: Mathematicians
   Task: Conduct research to extend mathematical knowledge in traditional areas, such as algebra, geometry, probability, and logic.

   Reward Rating: 2
   Risk Rating: 0
   Reward Justification: LLMs can speed up subtasks like proof sketching, counterexample search, and literature review with the expert validating all claims. However, existing systems struggle in coming up with the right ideas and sketches for frontier math problems.
   Risk Justification: The task is not safety critical, and therefore, there is no risk.

2. **Occupation**: Pharmacists
   Task: Provide information and advice regarding drug interactions, side effects, dosage, and proper medication storage.

Reward Rating: 3
Risk Rating: 5
Reward Justification: Best rewards are on communication and workflow (not core clinical judgment): draft patient-friendly counseling, generate clarifying questions and draft messages to prescribers, while the pharmacist verifies key facts in trusted databases.
Risk Justification: The task is safety critical as incorrect interaction, dose, side-effect, or storage advice can cause serious harm. LLMs raise risks mainly via confident clinical errors or omissions, and automation bias. Most errors are catchable if a pharmacist verifies using authoritative drug databases or labeling and uses the patient profile, but subtle mistakes and outdated guidance can slip through, causing substantial harm.

3. **Occupation**: Human Resources Managers
   Task: Identify staff vacancies and recruit, interview, and select applicants.

   Reward Rating: 4
   Reward Justification: high reward when used for operations, not decision-making: draft job ads or outreach, summarize resumes into structured fields (verified), generate interview guides or rubrics, schedule or communicate, and draft documentation..
   Risk Rating: 3
   Risk Justification: It's a critical task, and hiring errors can cause serious individual harm and create legal or reputational risk. LLMs raise risk via biased screening or interview framing, mishandling applicant PII, incorrect compliance guidance, and misuse. Many issues are catchable with structured rubrics, audits, and keeping LLM output advisory, not decisive, but subtle bias or privacy failures can slip through.

4. **Occupation**: Claims Adjusters, Examiners, and Investigators
   Task: Review police reports, medical treatment records, medical bills, or physical property damage to determine the extent of liability.

   Reward Rating: 4
   Risk Rating: 4
   Reward Justification: High reward for evidence organization: extract entities, build timelines, flag inconsistencies, draft letters, and generate follow-up questions.
   Risk Justification: The task is safety critical; wrong liability calls can seriously harm claimants or insurers and create legal or fraud exposure. LLMs raise risk by misreading or omitting evidence, hallucinating facts, making overconfident liability inferences, and introducing biased credibility judgments; sensitive medical or police data also heightens privacy or security risk, and fraudsters can tailor narratives to game the model. Humans can catch many issues if LLMs are limited to summarization or extraction and every claim is traced back to the record, but subtle distortions can slip through.